

Development of autonomous sailboat sails and future perspectives: A review

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ABSTRACT

Autonomous sailboats are a new type of long-endurance marine robots driven by marine renewable energy. They convert wind energy into driving force through sails. The aerodynamic performance and automatic control convenience of the sails significantly affect the sailing performance of autonomous sailboats. The sails of autonomous sailboats have evolved through three stages: traditional flexible sails, balanced rig with flexible sails, and rigid wingsails. Flap wingsails, self-trimming wingsails, and wingsails designed for specific conditions have been developed based on rigid wingsails. While autonomous sailboats typically use one sail, two or more sails can also be used. This review discusses the different configurations of autonomous sailboat sails, along with their advantages and disadvantages, key technologies, and research methods. It summarizes the features and technologies of autonomous sailboat sails that need further investigation, such as high strength, lightweight construction, multifunctional integration, interchangeable modularity, self-diagnostic intelligence, and deformable adaptability. Computational fluid dynamics simulations and wind tunnel tests will continue to be the primary methods for studying sail aerodynamic performance.

Nomenclature

Abbreviations	
AOA	Angle of attack
CFD	Computational fluid dynamics
FBG	Fiber Bragg grating
INNOC	Austrian Society for Innovative Computer Sciences
NACA	National Advisory Committee for Aeronautics
RANS	Reynolds-averaged Navier–Stokes
TOF	Time of flight
USNA	United States Naval Academy
USV	Unmanned surface vehicles
2D	Two-dimensional
3D	Three dimensional
Symbols	
V_A	Apparent wind speed (m/s)
C_l	Sail lift coefficient (–)
C_d	Sail drag coefficient (–)
C_{av}	Average value of wingsail chord
C_1	Wingsail tip chord
C_0	Wingsail root chord
C	Width of arc-shaped airfoil

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D	Wingsail span
ρ_a	Air density on the sea surface (kg/m ³)
S_A	Sail area (m ²)
S_w	Wingsail area of flap wingsail (m ²)
S_t	Tail area (m ²)
θ	Heeling angle (°)
δ_t	Flap deflection angle
T	Aspect ratio (–)
H	Outer arch ratio of arc-shaped wingsail (–)
E	Internal arch ratio of arc-shaped wingsail (–)
Σ	Camber ratio of arc-shaped wingsail (–)
λ	Taper ratio (–)
X	Sweep angle (°)
A	Sail angle of attack (°)
F_M	Driving force
F_H	Lateral force
F_L	Lift force on sail
F_D	Drag force on sail
f_1	Outer arch height
f_2	Inner arch height
L_w	Lift force of flap wingsail

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D_w	Drag force of flap wingsail
L_t	Lift force of tail
D_t	Drag force of tail
M_w	Wingsail additional torque (N/m)

1. Introduction

The increased growth requirements of air–sea interface observation and the progress of automatic control technology have promoted the development of autonomous sailboats. Autonomous sailboats convert wind energy into driving force through sails and harvest electrical energy from solar panels. Autonomous sailboats have been successfully used to overcome the energy-supply bottleneck and achieve long-endurance navigation. The sailboats carry various scientific payloads and perform long-term and large-scale ocean observation, providing a new observation method for the study of frontier scientific issues in the marine field [1], such as the observation of carbon flux at the air–sea interface [2] and monitoring temperature, wind, and solar radiation data in the Pacific Ocean [3].

Autonomous sailboats can be used to acquire real-time observation data in harsh sea conditions, which is difficult using traditional observation methods [4,5]. Compared with scientific research ships, the use of autonomous sailboats is economically advantageous for observing air–sea interactions. The CO₂ emission of traditional fuel-powered vessels [6] hinders the accurate measurement of the carbon flux at the sea surface; however, carbon footprints of autonomous sailboats are zero. An autonomous sailboat developed by Saildrone Company, equipped with scientific sensors, completed a circum-Southern Ocean voyage in 2019; this mission lasted 196 d and 11,879 nautical miles (Fig. 1a). In situ carbon flux data in the Southern Ocean was obtained; this data is crucial for the study of energy and material exchange at sea surfaces [2]. In situ observation of air–sea interface within the interior of a hurricane is valuable for studying hurricanes. The sea state inside hurricanes is rough, and traditional unmanned vehicles find it difficult to enter these hurricanes. In 2021, an autonomous sailboat named Saildrone passed through the eye of hurricane Sam successfully and obtained video images (Fig. 1b) as well as the air temperature, pressure, atmospheric humidity, wave height, and ocean current, providing data for a hurricane prediction model [7]. In 2022, two autonomous sailboats named TRITON were deployed near the Shushah Island, in the Red Sea, to collect and provide bathymetric data, video, and water chemistry sampling; these data are essential for investigating reefscape [8].

Autonomous sailboats comprise hull structure, sails, rudder, keel and bulb, photovoltaic power generation and energy management systems, sailing control system, path planning and navigation system, obstacle avoidance system, communication and position system, and scientific payloads. Yu et al. [9] introduced the technical status of autonomous sailboats and discussed the key technologies for their future development. Silva et al. [10] presented a survey of the developments in the

field of autonomous sailboats with rigid wingsails and described the primary hardware and software requirements. Fang et al. [11] summarized the technical status of traditional flexible sails from the perspectives of physical characteristics, installation, fixing methods, and sail adjustment. Sun et al. [12] investigated the existing design and control strategies for energy sufficiency of autonomous sailboats with respect to actuation, harvesting, and energy management. Tipsuwan et al. [13] classified the heading and speed control strategies for autonomous sailboats from the perspective of navigation control. Jing et al. [14] reviewed the challenges and related solutions of navigation of autonomous sailboats, including modeling analysis and path planning and implementation. These aforementioned studies provide valuable information for research on autonomous sailboats. Wind-assisted ship propulsion is significant for reducing fuel consumption and achieving green shipping in the present day. Rectangular sails, triangular sails, walker wingsails, rotor sails, and skysails are typically used in wind-assisted ships [15]. Despite the theory of using wind to provide driving force is the same, the types and technical routes of sails in the shipping industry differ from those of small autonomous sailboats. The primary differences are.

- 1) Large cargo ships are driven primarily by diesel engines and sails as assisted, whereas small autonomous sailboats are completely driven by sails.
- 2) Autonomous sailboats are more sensitive to the weight of sail compared to large wind-assisted ships, owing to their small displacement.
- 3) Autonomous sailboats are powered by solar panels. They are more sensitive to the power consumption of sail control compared to large wind-assisted ships.

Autonomous sailboat is a type of unmanned surface vehicle (USV); however, unlike diesel electric-powered unmanned boats, sails are the unique feature of autonomous sailboats. Fig. 2 illustrates the force on

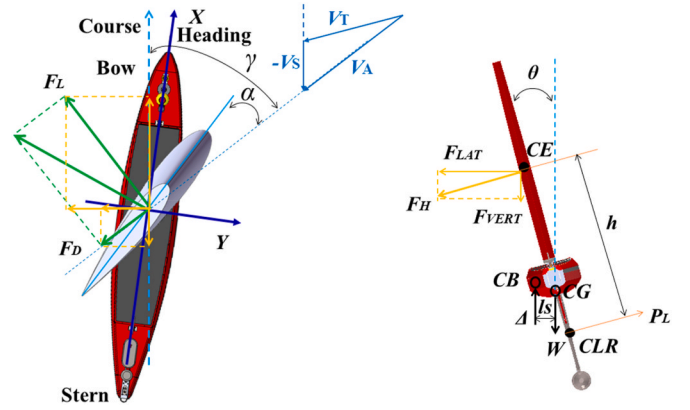
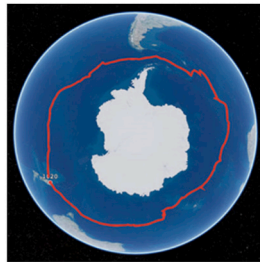
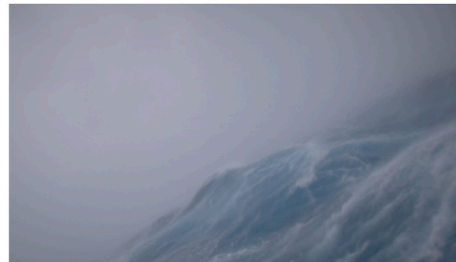


Fig. 2. Wingsail and hull force diagram.



(a) Saildrone observation in the Southern Ocean [9]



(b) Image of Hurricane Sam captured by Saildrone [10]

Fig. 1. Representative applications of Saildrone.

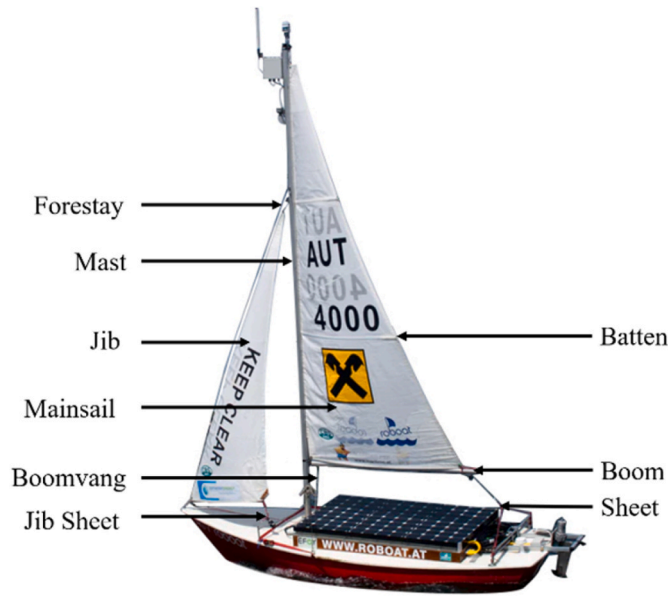


Fig. 3. Sail structure of ASV Roboat.

wingsail and hull of an autonomous sailboat. From Fig. 2 and Equations (1) and (2), the driving force F_M and lateral force F_H are respectively parallel and vertical to the course angle; they are the vector sum of the lift force on sail F_L and drag force F_D . Heeling and leeway angles are caused by F_H . Forces F_L and F_D are affected by the wind speed and aerodynamic parameters of the sails. To summarize, the influence of sails on autonomous sailboats are reflected in the following aspects.

- 1) The aerodynamic efficiency of sails affects the wind energy conversion efficiency and the maximum navigation speed of the sailboat.
- 2) The lift-to-drag ratio of sails affects the heeling and leeway angles of an autonomous sailboat.
- 3) Differences in the sheeting method of the sails affect the complexity of the sail-sheeting mechanism, power consumption, and convenience of automatic control.
- 4) The weight of sails affects the height of center of gravity of the autonomous sailboat, affecting the lateral stability of the sailboat.
- 5) The structural safety of sails is the key factor that affects the survival of the autonomous sailboat.

$$\begin{cases} F_L = \frac{1}{2} \rho_a \cdot V_A^2 \cdot S_A \cdot C_{l(\text{profile}, \alpha, Re, AR)} \\ F_D = \frac{1}{2} \rho_a \cdot V_A^2 \cdot S_A \cdot C_{d(\text{profile}, \alpha, Re, AR)} \end{cases} \quad (1)$$

$$\begin{bmatrix} F_M \\ F_H \end{bmatrix} = \begin{bmatrix} \sin(\gamma) & -\cos(\gamma) \\ \cos(\gamma) & \sin(\gamma) \end{bmatrix} \begin{bmatrix} F_L \\ F_D \end{bmatrix} \quad (2)$$

As sails are the core elements of autonomous sailboats, the study of sails is theoretically and practically valuable considering the technological development and extensive commercial application of autonomous sailboats. This review investigates three typical sail types used in autonomous sailboats: traditional flexible sails, balanced rig with flexible sails, and rigid wingsails. The structural characteristics, performance-influencing factors, advantages and disadvantages, and application scenarios of different sails are reviewed and the direction of development of autonomous sailboats sails is discussed. The remainder of this paper is structured as follows: Section 2 is focused on traditional flexible sails, and Section 3 presents a discussion of balanced rig with flexible sails. In Section 4, rigid wingsails are reviewed. Multi-sail layouts and new types of sails are discussed in Section 5. Section 6 presents a discussion of the topics covered in the previous sections and Section 7

Table 1

Autonomous sailboats use traditional flexible sails.

Platform name	Research institution	Reference (s)	Parameter(s)	Image
ASV Roboat	INNOC	[17,18]	Length: 3.7 m, Displacement: 300 kg, Sail area: 4.5 m ²	
Aeolus	Swiss Federal Institute of Technology Zurich (ETH)	[19,20]	Length: 1 m	
ASAROME	Pierre and Marie Curie University	[21]	Length: 3.6 m, Sail height: 2.25 m	
Nav2	Nav Ocean Company	[22]	Length: 2 m, Displacement: 38.5 kg, Draft: 0.75 m	
FASt	University of Porto	[23]	Length: 2.5 m, Displacement: 70 kg, Sail height: 3.4 m, Sail area: 3.7 m ²	
Raye	The University of British Columbia	[24,25]	Length: 5.5 m, Sail height: 6.1 m	
OceanVoy	The Chinese University of Hong Kong, Shenzhen	[26]	Length: 3.1 m, Displacement: 75 kg, Sail area: 3.75 m ²	

concludes the study.

2. Traditional flexible sails

2.1. Structure of traditional flexible sails

Advances in sailing yacht and automation technology effectively promote the development of autonomous sailboats. The initial autonomous sailboats were typically converted manned or remote-control sailboats used for the study of automatic navigation control or related teaching research [16]. Therefore, in the early stages of development, autonomous sailboats used the same or similar sails as manned sailboats. For example, the ASV Roboat autonomous sailboat (Fig. 3), developed by the Austrian Society for Innovative Computer Sciences (INNOC) uses traditional flexible sails, including cables, mast, jib, mainsail, boom, and

Table 2
Performance of traditional flexible sails in autonomous sailboats.

Advantages	Disadvantages
1 The shape and curvature of the sail can be changed by tensioning and releasing the ropes to suit different wind directions and conditions.	1 Through rope control, a risk of rope knotting exists that can cause the sailboat to lose control.
2 The sails are lightweight and low cost.	2 The sails are prone to distortion, resulting in different angles of attack at different positions on the sail and reduced navigation efficiency.
3 The sharp guide edge makes it insensitive to Reynolds number changes [35].	3 The lift-drag ratio is low (typically 3–5).
4 Maintenance is simple.	4 The shapes and control modes of flexible sails render automatic control inconvenient.

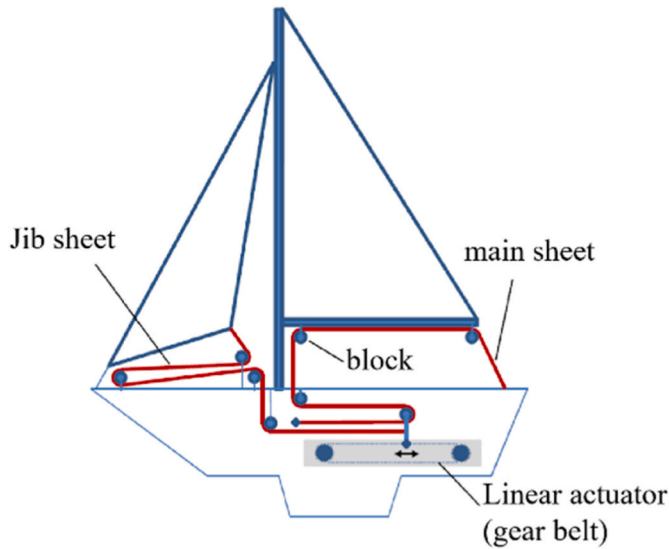


Fig. 4. Sail adjustment for the ASV Roboat autonomous sailboat [17].

boom vang. The jib, a triangular staysail, is used to improve handling and to increase sail area; this gives control over the bow of the sailboat, rendering it easier to maneuver the sailboat. The mainsail is attached to the boom behind the mast and runs to the stern. Representative autonomous sailboats with traditional flexible sails are listed in Table 1.

Luce Canon [27], USNA SailBot #3 [28], Sea Quester [29] and Mid-lenium Falcon [30] (developed by the United States Naval Academy (USNA)), MaxiMOOP (jointly developed by the USNA and the Aberystwyth university [31]), Roboat I (developed by Austrian society for Innovative Computer Sciences [32]), rrMM (developed by

University of Luebeck [33]), and Marius (developed by SUPMECA Toulon [34]) also use traditional flexible sails. The advantages and disadvantages of traditional flexible sails in autonomous sailboats are listed in Table 2.

2.2. Control mechanism of traditional flexible sails

The shape and curvature of traditional flexible sails are controlled by the jib sheet and mainsheet, providing optimal driving force under different wind directions. In sailing yacht, different sails are used according to various sailing conditions. For example, in the Volvo sailboat, eight sails can be brought on board during a voyage [36], the maximum and minimum sail areas are 375 m² and 29.7 m², respectively [37]. For autonomous sailboats, the sail area and type cannot be changed during a voyage although the sails' shape and curvature can be adjusted. When using traditional flexible sails on autonomous sailboats, different sailing conditions must be considered comprehensively when determining parameters such as the maximum sail area. The principle of traditional flexible sails adjustment in autonomous sailboats is the same as that in a sailing yacht. Autonomous sailboats typically use electric linear actuators (gear belts) and pulleys to pull the ropes and change the force direction of the jib sheet and mainsheet, achieving sail adjustment, as shown in Fig. 4.

2.3. Research methods of traditional flexible sails

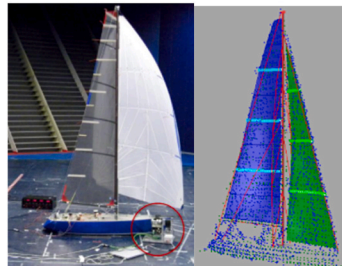
2.3.1. Technique of sail-shape capturing

Determining the flying shape of sails during sailing is the key to analyzing the changes in sail shape and study the influence of wind load on the flying shape [38]. This is crucial for sail performance optimization and evaluation. Methods to determine the flying shape include photogrammetry, time of flight (TOF) measurement, and fiber Bragg grating (FBG) measurement, as shown in Fig. 5.

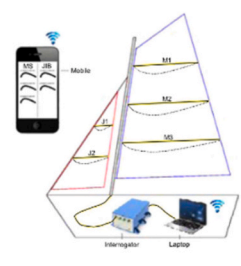
In photogrammetry, points are marked on the surface of a traditional flexible sail; a camera captures it at different angles. Then, coordinate analysis of the marked points is performed to invert the shape of the sail. Masuyama et al. [39] used this method for the full-scale measurement of the sail shape under upwind conditions; the sail shape and aerodynamic performance under different wind directions and wind speeds were measured, and a full-scale three-dimensional (3D) sail-shape database was established. To address the poor measurement results of the downwind sail state by the traditional sail-profile measurement methods, Deparday et al. [38] proposed a 3D curvature representation method. It displays the depth of the sail projected onto the planes formed by three angles; this system includes eleven load sensors, six cameras, and a wind direction sensor. Additionally, fifty-one markers were placed on the surface of each sail with all cameras deployed on the sailboat. This eliminated the need for an accompanying boat to assist with the measurement. However, this scheme is mainly aimed at static sails and is not successful for dynamic reconstruction of the sail.



(a) Photogrammetry [43]



(b) TOF measurement [44]



(c) FBG measurement [45]

Fig. 5. Flying-shape measurement methods.

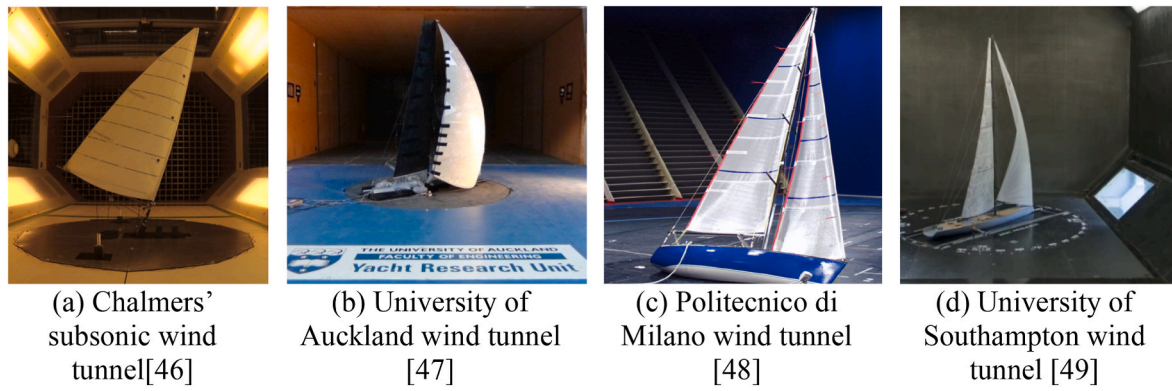


Fig. 6. Wind tunnel testing of sails.

Mausolf's [40] experiment showed that the sail shapes of the model in a wind tunnel and the full-size sailboat in a real lake were consistent; however, four cameras had to be arranged outside the ship and accompanying ships were needed. Maciel et al. [41] used the method of marking and positioning on the sail surface and realized the dynamic reconstruction of the sail shape through video recording from a single camera. The experiment was conducted under natural sunlight at sea; consequently, strong reflections led to poor results. The method also required an accompanying ship to carry the camera.

The TOF is a distance measurement technology. The distance between the transmitter and reflector is calculated by measuring the flight time of the ultrasonic, microwave, and light signals between them. This technology can provide depth information and can be combined with cameras to obtain 3D images. Fossati et al. [42] used a laser scanner measurement method based on TOF to determine the flying shape of sails during a voyage.

The FBG method involves arranging flexible beams on a

predetermined section of the sail at different heights and using sensors to measure the strain of the flexible beam. The strain information of the flexible beam was converted into the curvature information of the sail to reconstruct the flying shape. Ferreira et al. [43] used an FBG sensor to measure the flying shape of a sail. Ten FBG sensors were arranged in the fifth section of the sail and the flying shape was reconstructed based on the measurement results. The results were within an error range of 5 %.

2.3.2. Wind tunnel testing

Wind tunnel testing provides the benefit of a controlled environment, enabling accurate measurements of pressure and flying shape. As illustrated in Fig. 6, certain wind tunnels are specifically equipped for these measurements. Using data from wind tunnel tests, we can conduct detailed research on sail aerodynamics. This method has become a standard practice for investigating and optimizing sails for both upwind and downwind conditions.

Nicolas et al. [44] conducted an experiment in the University of Auckland yacht research unit's twisted flow wind tunnel to examine the effect of dynamic trimming on three mainsail models under upwind conditions, using a 1/13 scale IMOCA 60-foot design mainsail. Bayati et al. [45] described wind tunnel tests on sail aerodynamics in upwind conditions performed at Politecnico di Milano. Unlike tests limited to sails alone, this study utilized a complete 1:10 scale model of a 48' cruiser, including the yacht hull, deck, mast, rigging, and sails.

Magnander et al. [46] studied the ILCA 7 MKII sail in downwind conditions, presenting aerodynamic coefficients from wind tunnel tests for the ILCA 7 in all six degrees of freedom, with the sail scaled at 1/7. Campbell et al. [47] detailed tests conducted in four different wind tunnels (Volvo, Marchwood Engineering Laboratory, Politecnico di Milano, and University of Southampton) over a fifteen-year period from 1991 to 2007. This work highlighted the features of wind tunnels that affect results, noting that lift coefficients from different tests showed broad similarity within a 10 % tolerance band, while drag coefficients varied within 20 %.

Wind tunnel tests using scaled models can be affected by scale effects. To address this issue, Hansen et al. [48] conducted a comparison between wind tunnel and full-scale aerodynamic sail force measurements. They employed advanced wind tunnel testing techniques to more accurately replicate full-scale sailing conditions. With these enhanced methods, a generally good qualitative and quantitative agreement with the full-scale data was achieved.

2.3.3. Computational fluid dynamics simulation

Wind tunnel testing methods are limited by their long test periods, high pricing, and scale effects. They have now been partly replaced by computer simulation. An important parameter for predicting the performance of sailboat is the aerodynamic force on the sail surface. Masuyama [49] obtained sail-shape data using photogrammetry and used computational fluid dynamics (CFD) to simulate the flying shape;,

Table 3
Material characteristics of traditional flexible sails.

Materials	Advantages	Disadvantages	Typical applications
Polyester fiber	Low cost, water non-absorbent, good wear resistance, good Ultraviolet (UV) aging resistance	Unknown	
Nylon	Wear resistance, high tensile strength, low density, good dyeing ability	Poor resistance against chemical corrosion and UV aging	Jib
Polyester film	Good shape stability, lightweight	Poor stability under UV	
Laminated fabric			
Aramid fiber	Low density, long storage life, good resistance against chemical corrosion, high temperature resistance	Easy to absorb moisture, poor stability under UV	
Carbon fiber	High temperature resistance, good wear resistance, high tensile strength, low density	Costly	
High strength high-modulus polyethylene fiber	Low density, high tensile strength, good anti-fatigue property, good shock resistance	Pronounced creep, difficult to recover original shape	Jib
Liquid crystal Spinning	High strength, impact resistance, good wear resistance, good shock resistance	Poor UV resistance	Cruising mainsail

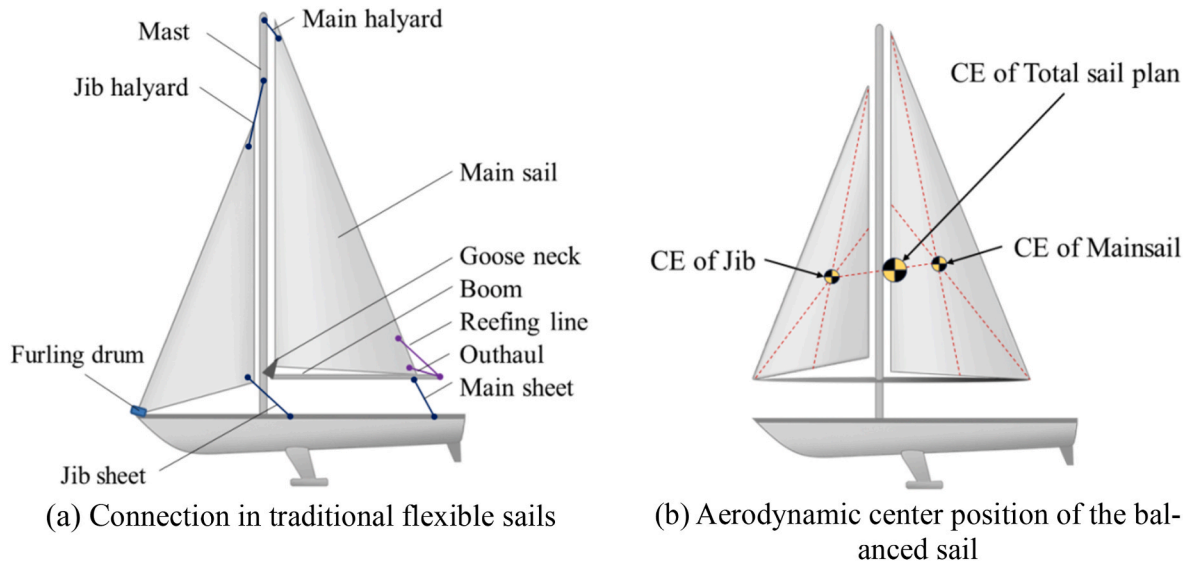


Fig. 7. Difference of between connections in a traditional flexible sail and balanced sail.

their simulation results agreed well with the measured data. Persson et al. [50] conducted wind tunnel experiments and used the Reynolds-averaged Navier–Stokes CFD method to investigate the steady and unsteady aerodynamics of the upwind sail. They found that the calculation results agreed well with experimental data under steady-state conditions, and the error was comparatively larger under unsteady-state conditions. Persson et al. [51] developed a time-domain model of unsteady sail aerodynamics based on the indicial response method; the prediction of driving and lateral force amplitudes were significantly improved.

2.4. Materials used in traditional flexible sails

The material of flexible sails in autonomous sailboats can be selected by referring to those of the traditional flexible sails in sailing yacht as reference. Cali et al. [52] compared the performance of sails using the eddy turbulence kinetic energy and aerodynamic coefficients of different canvas surfaces, providing reference information for the selection of sail materials. Cali et al. [53] explored the fluid–structure interaction of symmetric tri-radial sails, investigating the effects of fiber direction and sail layout on the flying shape and aerodynamic performance. Li et al. [54] reviewed commonly used textile materials for sails, and summarized the advantages and disadvantages of various fiber materials and their application scope. Laminated fabrics, made from multiple layers of fiber textiles bonded together with polyester films, are optimal traditional flexible sail materials. The characteristics of some traditional flexible sails materials are listed in Table 3.

3. Balanced rig with flexible sails

As shown in Fig. 7a and in conjunction with Fig. 3, traditional flexible sails feature a main head and mast connected through the main halyard. The outhaul binds the main clew and boom together, while the main tack connects via the gooseneck, and the boom attaches to the hull with the help of the mainsheet. The jib head and mast are linked by the jib halyard, with the jib sheet connecting to the jib clew and hull, and the jib tack usually connecting to the hull via a furling drum. In practice, the connection method is more complex than described. Sail shape is adjusted by trimming or easing the mainsheet or jib sheet, a process that can be complex.

To simplify sail-shape control and enhance usability, a balanced rig with flexible sails (referred to as balanced sails) was developed [55], as shown in Fig. 7b. In this setup, the mainsail and jib are connected only to

Table 4

Autonomous sailboats that use balanced sails.

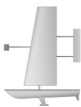
Platform name	Research institution	Reference (s)	Parameter(s)	Image
Vaimos	French Research Institute for Exploitation of the Sea (IFREMER)	[61]	Length: 4 m	
Iboat II	University of Toulouse	[62,63]	Length: 2.4 m, Displacement: 35 kg, Sail area: 1.5 m ²	
Avalon	ETH	[57]	Length: 3.95 m, Sail height: 5.7 m, Sail area: 8.4 m ²	

the mast and boom, with no direct connection to the hull. This separation reduces the risk of out-of-control situations caused by rope winding and knotting. The mast can rotate around its center of rotation. By adjusting the length of the main boom (the front boom of the mast) and the auxiliary arm (the rear boom of the mast), the force on the mainsail can be made slightly greater than that on the jib. This adjustment positions the center of effort (CE) of the total sail plan closer to the mast, achieving a balanced air force between the jib and mainsail around the mast (Fig. 7b).

Compared with traditional flexible sails, the key difference of balanced sails is that the mainsail and jib are no longer connected to the hull, and the mast can be rotated. The following are a few advantages.

- 1). **Few fixed points:** Balanced sails are completely separated from the hull and do not require sheets and pulleys. The simple structure increases reliability.

Table 5
Classification of rigid wingsails used in autonomous sailboat.

	Traditional wingsails	Flap wingsails	Self-trimming wingsails
Configuration			
Brief structural description	Single wingsail	Single wingsail with flap	Single wingsail with tail(s)
Controlled object	Wingsail mast	Wingsail mast, flap rotation shaft	Flap rotation shaft
Wing sail rotation driving method	Active rotation	Active rotation	Passive rotation
Structural complexity (including driving mechanism)	High	Middle	Low
Control complexity	Low	High	Middle
Energy consumption	High	Middle	Low
Precise control of AOA	High	High	Low
Others		High lift-to-drag ratio	Poor driving force when sailing downwind

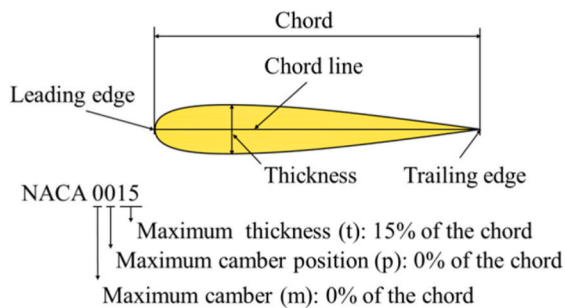
- 2). **Easy automatic control:** Traditional flexible sails are controlled by a rope and pulley system, whereas balanced sails use a motor to rotate the mast. The latter setup is more convenient for automatic control [56].
- 3). **Low control power consumption:** Balanced sails reduce the power required to adjust the sails by 50 % [57], effectively decreasing the control energy consumption.
- 4). **High aerodynamic efficiency:** The lower boom of a balanced sail helps guide the wind across the sail's surface, increasing the lift coefficient and lift-to-drag ratio. As a result, a balanced sail can provide greater driving force and produce less lateral force compared to traditional flexible sails [58].

As the main body of a balanced sail is made of flexible materials, their disadvantages are the same as those of the traditional flexible sails.

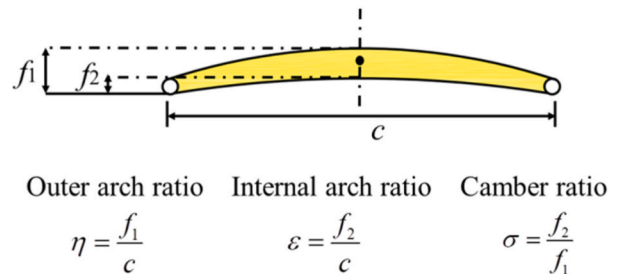
Table 4 lists the autonomous sailboats that use balanced sails (apart from the Breizh Spirit autonomous sailboat developed by the University of Western Brittany [59] and the autonomous sailboats developed by the USNA [60]).

4. Rigid wingsails

Rigid wingsail is a new type of sail designed with reference to aircraft wings [10]. It is the representative sail in autonomous sailboats owing to its advantages such as good aerodynamic stability, high aerodynamic efficiency (maximum lift coefficient of traditional flexible sails is 0.8



(a) Symmetrical airfoil



(b) Arc-shaped airfoil

Fig. 8. Parametric expressions of traditional wingsails.

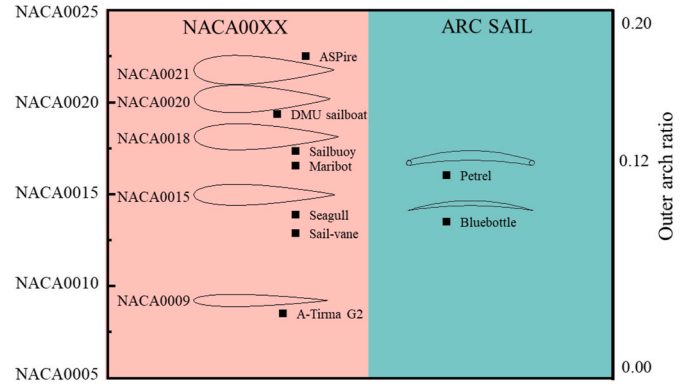


Fig. 9. Various wingsail airfoils of autonomous sailboat.

whereas that of rigid wingsails is 1.15 [64]), and ease of automatic control [65].

Rigid wingsails, like balanced rig with flexible sails, rotate around the mast to maintain an optimal angle of attack (AOA). They can be classified into traditional wingsails, flap wingsails, self-trimming wingsails, and wingsails designed for special conditions, based on their structure, control method, and operational conditions. The first three types are summarized in Table 5. Traditional wingsails are the basic form of rigid wingsails, where a motor connected to the mast drives the wingsail's rotation. Flap wingsails, an advancement over traditional wingsails, also rotate similarly, but they feature adjustable flap deflection angles. This adjustment alters the curvature of the wingsail to enhance the lift coefficient and upwind sailing performance. Self-trimming wingsails differ in that their rotation is not directly driven by a motor. Instead, the tail functions as an air rudder; changing the deflection angle of the tail passively controls the AOA of the wingsail. This approach reduces energy consumption, which is particularly beneficial for autonomous sailboats with limited energy resources.

4.1. Traditional wingsails

The main factors affecting the aerodynamic performance of traditional wingsails are same as that in aircraft wings: sail area, aspect ratio, taper ratio, sweep angle, and airfoil section.

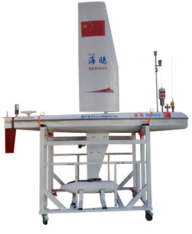
4.1.1. Traditional wingsail area

Equation (1) indicates that the lift and drag of the wingsail are proportional to the sail area S_A . The driving force trends to increase as the sail area increases. However, the heeling moment and lateral force also increase as the sail area increases. The heeling and leeway angles as well as the driving force should be considered when determining the sail area.

Generally, the area of a rigid wingsail cannot be changed. The

Table 6

Autonomous sailboats use traditional wingsails.

Platform name	Research institution	Reference	Parameters	Image
Triton	OCEAN AERO	[67]	Length: 4 m, Beam: 1.5 m, Water line to top of sail: 3 m, Water line to keel bottom: 1.5 m, Displacement: 350 kg	
Sailbuoy	Offshore Sensing AS	[68]	Length: 2 m, Beam: 0.52 m, Water line to keel bottom: 0.3 m, Water line to top of sail: 1.13 m, Displacement: 60 kg, Sail area: 0.3 m ² , Airfoil: NACA0018	
Seagull	Shenyang Institute of Automation Chinese Academy of Sciences	[69]	Length: 4 m, Airfoil: NACA0015, Sail height: 2.7 m, Sail area: 2.2m ²	

maximum sail area that can be used must be comprehensively determined considering the maximum wind speed and the transverse stability curve of the autonomous sailboat. This prevents the autonomous sailboat from capsizing and ensures navigation safety.

4.1.2. Traditional wingsail airfoil

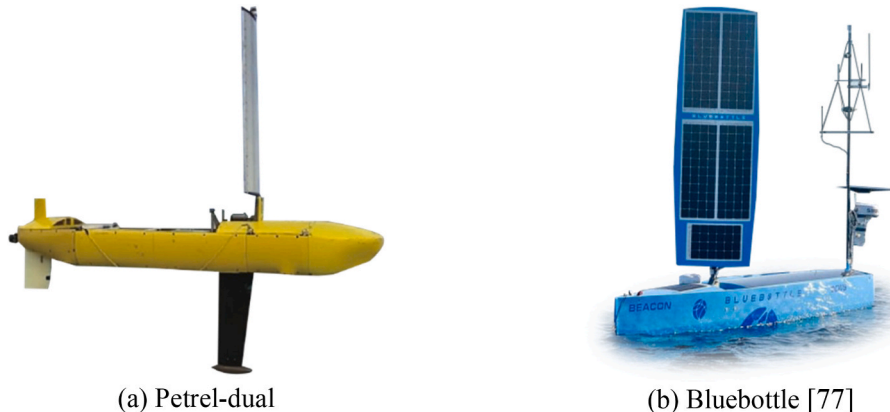
Traditional wingsails include symmetrical airfoil wingsails and arc-shaped wingsails. These differ in their airfoil profile. Fig. 8 illustrates the two typical traditional wingsail airfoil parameters. The airfoils of some autonomous sailboat are shown in Fig. 9.

4.1.2.1. Traditional wingsail with symmetrical airfoil. Unlike aircraft wings that have a certain degree of camber to generate lift, the airfoil of an autonomous sailboat wingsail is generally symmetrical. This is because autonomous sailboat must sailing in the wind from the port and starboard sides. Many high performance airfoils were developed by the Royal Aircraft Establishment and National Advisory Committee for Aeronautics (NACA). NACA four-digit airfoils and five-digit airfoils are typical low speed airfoils. Four-digit airfoils have high lift coefficients

and low drag coefficients, making them suitable for low speed vehicles such as autonomous sailboats. The autonomous sailboats listed in Table 6 adopt symmetrical airfoil wingsails. Ma et al. [66] analyzed the lift and drag coefficients, lift-to-drag ratio, and pressure contours of NACA2312, NACA0012, and NACA0018 airfoils at different wind speeds and AOA to identify suitable sails for autonomous sailboats.

4.1.2.2. Traditional wingsail with arc-shaped airfoil. Yin et al. [70] analyzed the influence of coupling parameters on the aerodynamic performance by conducting CFD simulation and orthogonal analysis, noting that the camber ratio is the core influencing factor. A higher camber ratio is beneficial for improving the downwind and upwind thrust coefficients; however, it also causes an increase in the lateral force. The outer arch ratio has the least influence on the aerodynamics performance of the arc-shaped wingsail; a lower outer arch ratio indicates a higher upwind thrust coefficient and lower upwind lateral force coefficient.

Arc-shaped wingsail and wingsail with NACA0015 airfoil were analyzed and compared in terms of the wingsail load, flow field, and

**Fig. 10.** Autonomous sailboats with arc-shaped wingsails.

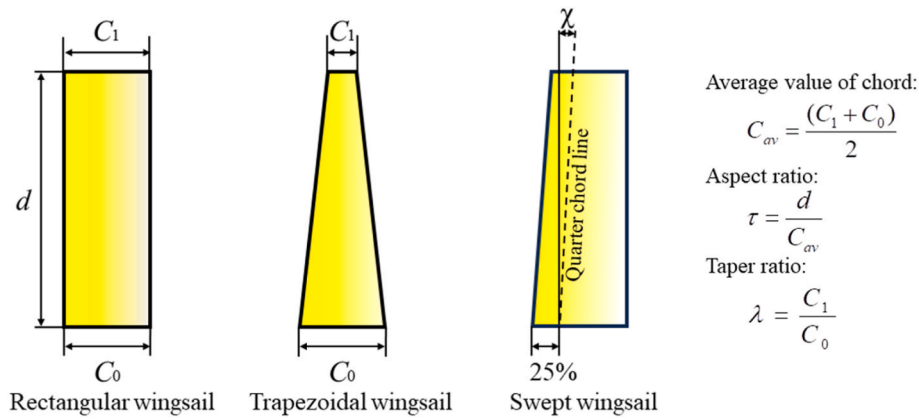


Fig. 11. Three-dimensional configurations of wingsails.

propulsion performance [71]. The results indicated that the thrust of the arc-shaped wingsail was higher than that of the wingsail with NACA0015 airfoil owing to the side edge vortex on the side of the arc-shaped wingsail; however, the arc-shaped wingsail exhibited a stronger flow separation phenomenon.

The Petrel autonomous sailboat (Fig. 10a) developed by Tianjin University and the Bluebottle autonomous sailboat (Fig. 10b) developed by Ocius Technology of Australia both use arc-shaped wingsails.

4.1.3. 3D configuration of traditional wingsails

The aspect ratio, taper ratio, and sweep angle of wingsails are shown in Fig. 11. Li et al. [72] observed that the wingsail lift coefficient increased with the aspect ratio while the drag coefficient decreased when the sail area and AOA were the same. Considering the Seagull autonomous sailboat as the research object, Sun [73] determined that the maximum sailing speed in different headings increased with the aspect ratio. However, the heeling moment caused by the wind also increased, which reduced the survivability of the boat in rough seas.

In windward state, as the sweep angle increases, the drag of the wingsail and the aerodynamic center of the wingsail decrease. However, the increase in the sweep angle resulted in a decrease in the lift coefficient.

Increasing the taper ratio can reduce the bending load at the root of the wingsail and decrease the height of wingsail center of gravity; this is beneficial in improving the structural reliability of the wingsail.

4.1.4. Mechanical structure of traditional wingsails

The mechanical structure of a wingsail is similar to that of an aircraft wing and comprises an internal rib, spar, and outer skin. In the wingsail, the mast has the same function as the main spar in an aircraft wing. The aerodynamic center of the wingsail exists at approximately quarter chord, which is the location on the wingsail where the moment coefficient is independent of AOA. Therefore, the mast is located at or near the 25 % chord location. The internal rib and mast are conventionally made of metal, and the outer skin is made of carbon fiber or polyurethane film. Banks et al. [74] conducted a wind tunnel test and CFD simulation to analyze the effects of sail skin deformation under the wind force on parameters such as the lift-to-drag ratio of the wingsail, they used a digital image correlation system to measure the deformation of the sail skin; the results revealed that when the wingsail skin was blown by the wind, approximately 15 % depression was generated, resulting in a 5%–11 % reduction in the lift-to-drag ratio.

Niu et al. [75] studied the influencing factors and laws of stress at the root of the wingsail mast. Their results indicated that the mast primarily bearing torsional loads, with the bending moment having minimal effect on the mast. The findings of this study provide reference for the structural design of wingsails.

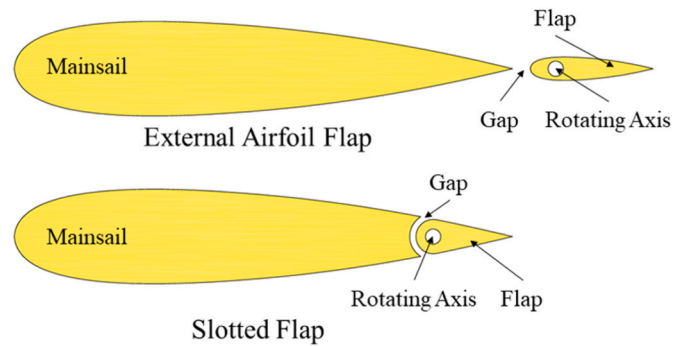


Fig. 12. Schematic of external airfoil flap and slotted flap.

4.2. Flap wingsails

The aerodynamic performance of a wingsail can be effectively improved by adding flaps or using multi-element wingsails based on symmetrical airfoils as the lift coefficient of the symmetrical airfoil is relatively low. The flap wingsail can adjust the camber of the entire wingsail by deflecting the flap, thereby increasing the lift coefficient and improving the wind-energy conversion efficiency. A significant advantage of flap wingsails over traditional wingsails is their ability to generate greater driving force with the same area, especially during the upwind sailing stage. Flap wingsails can be categorized into external and slotted flaps, as shown in Fig. 12. Details of autonomous sailboats equipped with flap wingsails are listed in Table 7.

The factors influencing the aerodynamic performance of flap wingsail include the arrangement of the flap (external airfoil flap or slotted flap), position of the flap relative to the mainsail, thickness of the flap, gap size, deflection angle of the flap, and chord length ratio of the mainsail and flap.

Li et al. [80] studied the aerodynamic performance of a flap wingsail, the results showed that compared with slotted flap, the external airfoil flap increased the rotation torque of the wingsail. The performance of the slotted flap was superior to that of the external airfoil flap in terms of lift-to-drag ratio and lift coefficient. The upward and downward movement of the flap position did not affect the lift coefficient, and the closer the flap was to the main sail, the smaller the lift coefficient of the flap wingsail. As the deflection angle of the flap increased, the lift coefficient, lift-to-drag ratio, thrust coefficient, and corresponding lateral force coefficient increased. However, this study primarily used two-dimensional (2D) fluid analysis, and there could be some deviations when it is extended to 3D.

Blakeley et al. [81] conducted a CFD simulation and wind tunnel test to study the effects of the gap size on the aerodynamic performance of a

Table 7

Autonomous sailboats using flap wingsails.

Platform name	Research institution	Reference(s)	Parameters	Image
Datamaran (mark7)	Autonomous Marine Systems	[76,77]	Length: 3.7 m, Water line to top of sail: 3.1 m	
Datamaran (mark8)	Autonomous Marine Systems	[77]	Length: 5 m, Water line to top of sail: 4 m	
Triton	OCEAN AERO	[78]	Length: 4 m, Beam: 1.5 m, Water line to keel bottom: 1.5 m, Displacement: 350 kg	
Saildrone (Hurricane Wing)	Saildrone	[79]	Length: 7 m, Sail height: 3 m	

flap wingsail. The results revealed that when compared to a gap size in the range of 0–2%, a larger gap size caused the wingsail to stall earlier.

Wang et al. [82] explored the influence of the gap size of slotted flap on the aerodynamic performance, and the results indicated that the lift and drag coefficient of the flap wingsail increased with an increase in the flap deflection angle, while the lift coefficient decreased as the gap size increased.

The 2D simulation results showed that when the rotating axis was located at 85 % of the chord direction of the mainsail, the stall was delayed with the thickening of the flap, whereas when the rotating axis of the flap was located at 75 % of the chord direction of the mainsail, the stall was delayed as the flap deflection angle increased at a lower flap deflection angle. The 3D simulation showed that when the flap deflection angle was low and the wingsail had stalled, the lift coefficient of the wingsail increased and then decreased as the flap rotating axis moved backward in the chord direction of the mainsail. When the wingsail was not stalled, changing the chord position of the rotating axis of the flap in the mainsail had little effect on the lift coefficient. Further, when the flap deflection angle was large, the lift coefficient of the wingsail increased with the rotating axis of the flap moving backward in the chord direction of the mainsail [83].

Kuang et al. [84] performed numerical simulation and indicated that an increase in the chord length ratio of the mainsail to the flap was beneficial for the increase in the thrust coefficient and lift coefficient of the flap wingsail at a given AOA. The lift coefficient of the flap wingsail increased with an increase in the deflection angle of the flap, and the optimal chord length ratios of the mainsail to the flap were 1.5–3.

In summary, the shortcomings of the existing flap wingsail study are as follows.

- 1). The current study mainly focuses on the effects of the position of the flap, arrangement of the flap, gap size, and other factors on the aerodynamic performance of the wingsail. Few studies have been conducted on factors such as the area ratio of the flap to the mainsail and the thickness ratio.

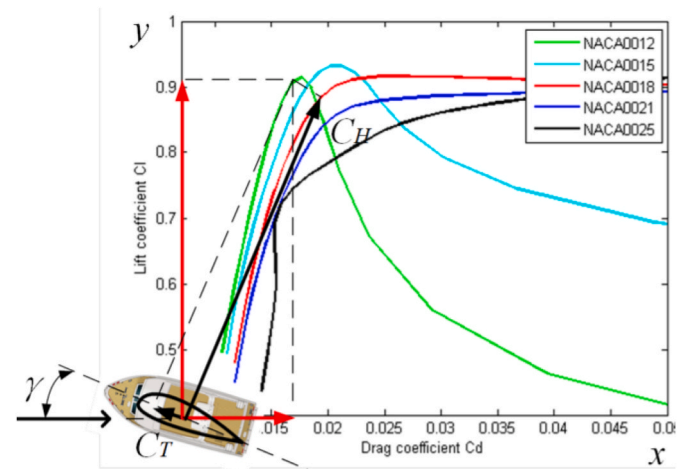


Fig. 13. Autonomous sailboat propulsion coefficient polar curves with different wingsail airfoils.

- 2). Previous studies have mostly involved 2D simulations, and the aerodynamic characteristics of 3D airfoil should be further studied.
- 3). Most studies have only focused on the effects of a single variable on the aerodynamic performance, and the coupling between different factors was ignored.

4.3. Self-trimming wingsails

In addition to the airfoil, camber, and other configuration parameters, the angle of attack is a key factor that affects the aerodynamic performance of a wingsail. The lift coefficient C_l and drag coefficient C_d are closely related to the AOA. To improve the wingsail wind energy conversion efficiency, the rotation of the wingsail should be controlled according to the changes in the course angle and wind direction to

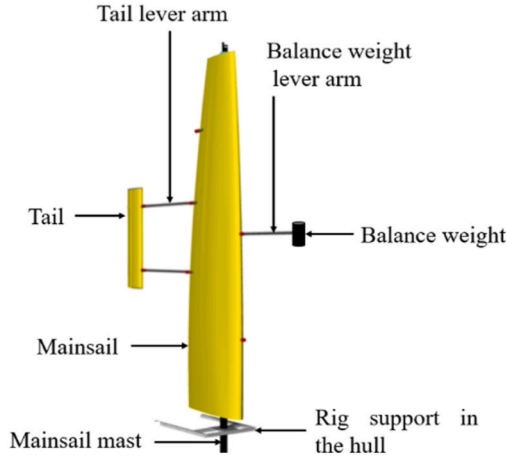


Fig. 14. Self-trimming wingsail structure and working state diagram (modified from Ref. [87]).

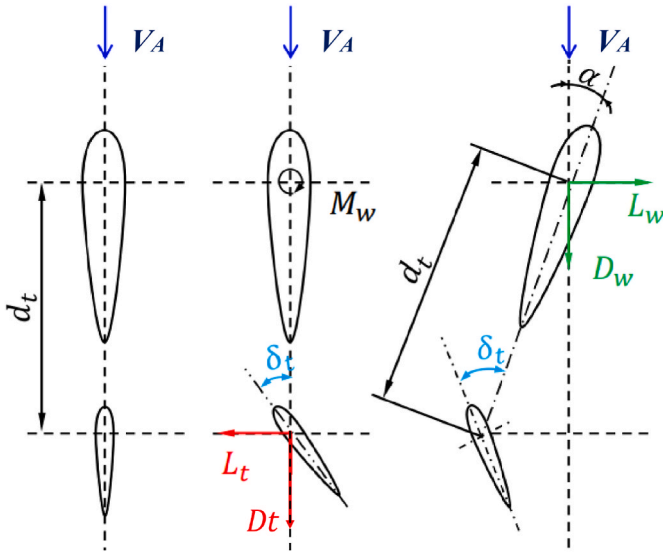


Fig. 15. Moment balance relationship of self-trimming wingsail.

maintain optimal AOA. In particular, during upwind sailing, when the wingsail deviates from the optimal AOA (the peak position in Fig. 13 corresponds to the AOA), the lift coefficient decreases rapidly, resulting in a rapid decrease in the driving force. He et al. [85] determined the coefficient of performance and optimal AOA of a sail model by performing a wind tunnel test. Huang et al. [86] conducted a study on the AOA of the sail to obtain the maximum speed of an autonomous sailboat.

As shown in Table 5 in Section 4, the rotation of traditional wingsails and flap wingsails are driven by motors; however, this causes the power consumption to be high. Moreover, a strong coupling exists between the wingsail and hull, and the yawing motion of the autonomous sailboat causes the wingsail to deviate from the optimal AOA. The structure of the self-trimming wingsail is shown in Fig. 14. It is composed of the mainsail, tail lever arm, tail, and balance weight. The tail lever arm connects the mainsail and tail, and the balance weight lever arm connects the mainsail and balance weight. The mainsail can freely rotate around the mast.

The tail is capable of active adjustment and acts as the “air rudder” of the mainsail. The mainsail is no longer driven by motor; however, by controlling the deflection angle of the tail relative to the mainsail, according to the principle of moment balance around the mainsail mast, the AOA of the mainsail can be passively controlled, and the mainsail will be in a passive stability state as shown in Fig. 15 and Equation (3).

Owing to the small tail area, the deflection angle can be adjusted quickly. Owing to the leverage effect, the torque required for tail adjustment is small, which can effectively reduce the power consumption. A self-trimming wingsail realizes decoupling of the mainsail and hull, and the wingsail does not oscillate with the wave together with the hull [88].

$$M_w + \frac{1}{2} \rho V_A^2 C_{Lw} S_w - \sum_{i=1}^{2n} \frac{1}{2} \rho V_A^2 C_{Lt} S_t d_t = 0 \quad (3)$$

Among sailboats equipped with self-trimming wingsails, the Sail-drone autonomous sailboat is the most representative. Table 8 highlights other notable sailboats with self-trimming wingsails, including the SailVane autonomous sailboat [89] and the Trimaran developed by the Royal Military College of Canada [90] both of which also utilize self-trimming wingsails.

The tails of self-trimming wingsails generally arranged at the horizontal position of the aerodynamic center of the mainsail. Some self-trimming wingsails have tails arranged at higher positions or at the top of the mainsail, as shown in Fig. 16. A tail with a higher position can take advantage of the wind at a higher speed, far away from the sea surface. This means that a smaller tail area is needed to control the mainsail. However, a higher position of the tail also leads to a higher center of gravity of the entire sail and increases the heeling moment. Therefore, self-trimming wingsails with the tail arranged at a high position install the front balance weight at a low position.

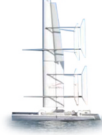
The significant advantage of self-trimming wingsail is that the AOA of the mainsail can be adjusted rapidly. However, the mainsail can only work at a small AOA (below the stall angle), which results in low driving forces when sailing downwind. According to the autonomous sailboat velocity polar curve shown in Fig. 17a, the speed of an autonomous sailboat with self-trimming wingsail is lower than that of an autonomous sailboat with active adjustment sail when sailing downwind. This is because when an autonomous sailboat sailing downwind, the optimal wingsail AOA is larger than the stall angle (Fig. 17b), and the drag force is the main component of the driving force. However, the tail cannot turn the mainsail to the required optimal AOA when sailing downwind.

The static and dynamic stabilities are the core performance parameters of self-trimming wingsails. The former determines the seaworthiness range and driving force, and the latter determines the oscillation time and the number of oscillations. The main factors influencing the performance of a self-trimming wingsail are the mainsail-tail area ratio and relative position of the mainsail-tail.

Elkaim [64] analyzed the aerodynamic performance at low Reynolds numbers based on the Atlantis project and evaluated the stability and sweep radius of four different configurations of self-trimming wingsails. Setiawan et al. [98] and Ulysse et al. [99] conducted an experimental

Table 8

Autonomous sailboats using self-trimming wingsails.

Platform name	Research institution	Reference (s)	Parameters	Image
Saildrone Explorer	Saildrone Company		Length: 7 m, Sail height: 5 m	
HWT X-1	Harbor Wing Technologies	[91]	Length: 9.1 m, Sail height: 10.7 m	
HWT X-3	Harbor Wing Technologies	[92]	Length: 15.4 m, Sail area: 74 m ²	
ASPIre	Aland University of Applied Sciences	[93,94]	Length: 4.18 m, Mainsail height: 2.8 m, Chord length: 0.74 m, Airfoil: NACA632-618, Tail height: 0.8 m, Tail chord length: 0.3 m, Material: carbon fiber	
Open Transatlantic Boat	OpenTransat	[95]	Length: 2 m, Sail height: 1.3 m	
Maribot Vane	KTH Royal Institute of Technology	[87]	Mainsail area: 2.7 m ² , Tail area: 0.3 m ² . Mainsail height: 3.5 m, Mainsail root chord length: 0.5 m, Top chord length: 0.45 m. Airfoils: both NACA0018	



(a) Maribot



(b) Walker Wingsail

Fig. 16. Some self-trimming wingsail tails are arranged high, and the balance weight is placed low.

analysis to study the performance of a self-trimming wingsail at different tail deflection angles. Tretow used the vortex lattice method to model and analyze a self-trimming wingsail, the analysis revealed that the tail should be kept far from the mainsail to reduce impact from mainsail on the lift coefficient of tail overall. However, this method only considers the induced drag, does not include the viscous drag, and is limited to the case of a small AOA. Zhao [100] analyzed the effects from mainsail and

tail on the aerodynamic performance and conducted lightweight design of the skin, wing beam, and rib of the mainsail.

Setiawan et al. [101] established a four-degree-of-freedom autonomous sailboat dynamics simulation model and analyzed the influence of the deflection angle of the tail and the rudder on ship dynamics, revealed the basic characteristics of sail-ship dynamics. Augenstein et al. [102] proved that the tail can not only adjust the AOA of the mainsail, but also

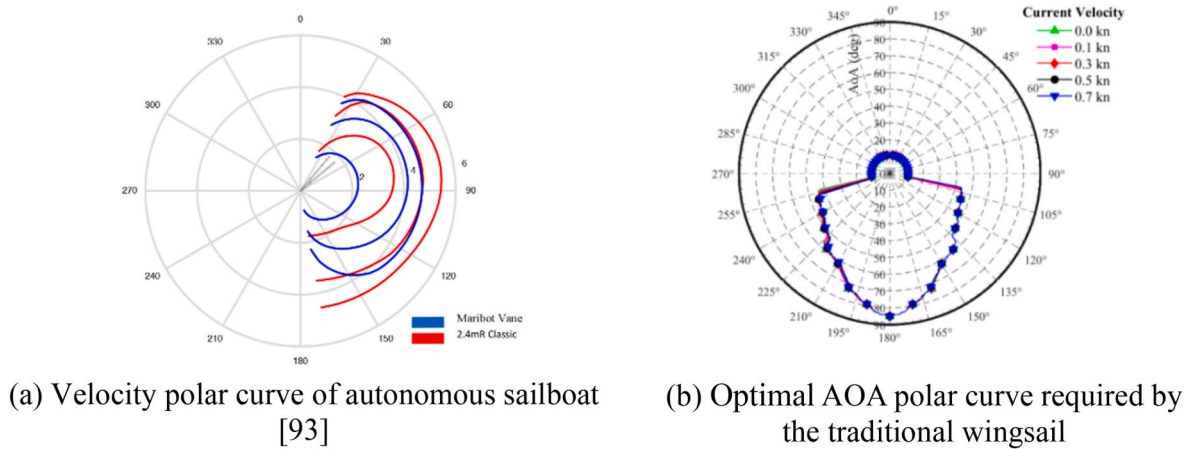


Fig. 17. Velocity polar curve and AOA prediction of autonomous sailboat wingsail.



Fig. 18. Saildrone Explorer SD1020, SD1022, and SD1023 autonomous sailboats operating in Antarctic region.

play a positive role in the course dynamic stability of autonomous sailboat.

The tail area generally accounts for 12%–15 % of the mainsail area. The tail area increases the moment of inertia of the self-trimming wingsail, causing an increase in oscillation time and a decrease in stability. The increase of tail area leads to a decrease in lift coefficient of the mainsail and an increase in the lift coefficient of the tail. Offshore Sensing Company [103] and Wang [104] also conducted research on self-trimming wingsails with elastic rope constraints, which provided a new idea for angle adjustment of wingsails.

4.4. Wingsails under special conditions

Although the sails of autonomous sailboats cannot be changed when sailing, they can be designed to adapt to different sea conditions in order to satisfy the requirements of wingsail structure safety, navigation driving force, and navigation safety under special working conditions. The wingsails for special working conditions can be categorized into polar wingsails, hurricane wingsails, super-large wingsails, and foldable wingsails.

4.4.1. Polar wingsails

The fifth-generation Saildrone autonomous sailboat, which is used for Antarctic navigation, initially used a rigid wingsail with a height of 15 feet. This enabled it to sail at wind speeds of up to 60 kn. However, this narrow and high wingsail could not withstand the extreme sea conditions in Antarctic region. In 2015 and 2017, the Saildrone Explorer autonomous sailboat was deployed twice in the Southern Ocean to sail around the Antarctic; however, the high-aspect-ratio wingsail was damaged on both voyages, as shown in Fig. 18a. Therefore, the Saildrone team designed a square wingsail fixed to the hull of the boat, as



Fig. 19. SD1048 Saildrone Explorer autonomous sailboat equipped with hurricane wingsail [107].

shown in Fig. 18b, and this remained intact even after the sailboat was capsized by a wave 50 feet high.

In subsequent sea trials, three autonomous sailboats numbered SD1022, SD1023, and SD1020 were deployed. The first two were equipped with enhanced self-trimming wingsails and were damaged during the first few days of the experiment. SD1020 was equipped with square wingsails, and the square wingsail remained intact during the storm. SD1022 and SD1023 successfully crossed the Drake Strait in winter after refitting the square wingsails (Fig. 18c) [105].



Fig. 20. Saildrone Surveyor autonomous sailboat [108].

4.4.2. Hurricane wingsails

In 2021, the National Oceanic and Atmospheric Administration (NOAA) and Saildrone company collaborated to deploy 24 Explorer autonomous sailboats equipped with hurricane wingsails during tropical storms to measure various parameters of the ocean surface and lower atmosphere [106]. Hurricane wingsails can cope with extreme wind and wave conditions, as shown in Fig. 19.

To enter hurricanes, the wingsail of Saildrone was shortened from a standard height of 5 m–3 m [109]. This enabled the hurricane wingsail to have a smaller area, making it capable of withstanding the impact of high-speed winds and waves at the eye of a hurricane (see Fig. 20). A smaller wingsail produces a smaller heeling moment, which can provide sailboats with strong heeling stability in large wind and wave conditions. Although the sail area is reduced, the hurricane wingsail increases the camber of the wingsail through the flap to maintain sufficient power. Unlike the flap wingsails described in Section 4.2, the flap of a hurricane wingsail exists only in the upper half of the wingsail. This design reduces the probability of the flap being hit by waves and effectively increases the reliability. In 2021, the autonomous sailboat numbered SD1045 sailed into the eye of hurricane Sam, and the observed maximum wind speed was up to 109.83 kn. In 2022, autonomous sailboat numbered SD1078 sailed at a speed of more than 7.8 kn in a Category 4 storm, with a peak speed of 34.5 kn and a maximum wave height of 17 m [7].

4.4.3. Super-large wingsails

In addition to Explorer (having a length of 7 m) and Voyager (having a length of 10 m), Saildrone also developed Surveyor, which had a length of 20 m and sail height of 13 m. Using an electric motor for wingsail rotation results in significant energy consumption. Therefore, the self-trimming wingsail is the first choice for a super-large wingsail.

In March 2024, the construction of the latest model of Saildrone Surveyor autonomous sailboat was completed [110]. The wingsail was made of fiberglass, and the hull was made of aluminum. Compared with

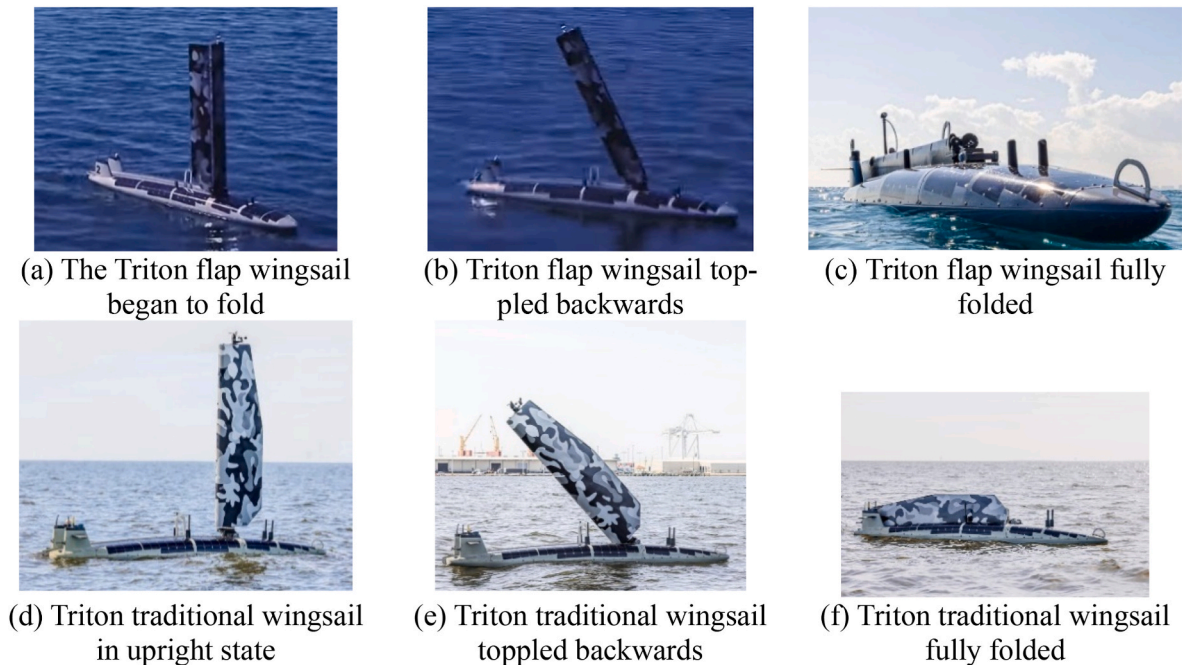


Fig. 21. Comparison of Triton wingsail before and after folding [111].

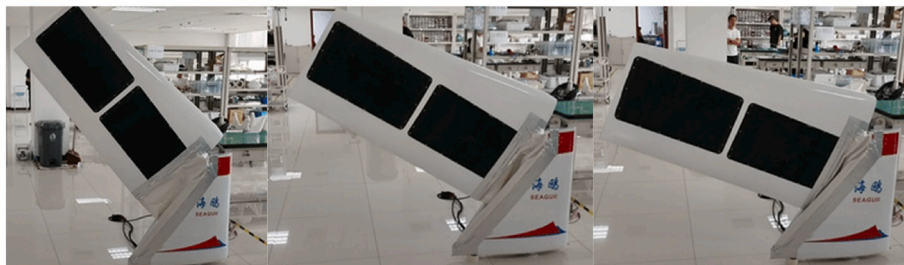
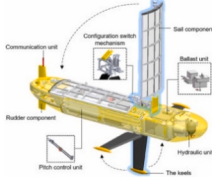


Fig. 22. Seagull autonomous sailboat foldable wingsail.

Table 9
Autonomous sailboats with backward-folding, arc-shaped wingsails.

Platform name	Research institution	Reference	Image
Bluebottle	Ocius Technology	[96]	
Petrel-dual	Tianjin University	[97]	

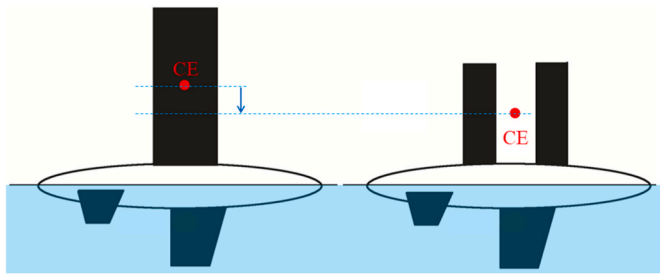


Fig. 23. Force schematics on single and dual sails (modified from Ref. [112]).

previous products, the new wingsail is thinner and the tail length is shorter. These designs effectively reduce the weight of the wingsail.

4.4.4. Foldable wingsails

Ocean Aero launched the Triton autonomous sailboat (Fig. 21), an autonomous underwater and surface vehicle that can fold its wingsail and dive underwater to avoid bad weather. Triton uses two types of wingsails: foldable flap wingsail and foldable traditional wingsail. For the foldable flap wingsail, the flap is folded first (Fig. 21a) and then the wingsail is laid down backward into the groove on deck (Fig. 21b and c). For the traditional wingsail, first the leading edge of the wingsail is turned toward the stern of the boat (Fig. 21d) and it is then laid down backward into the groove on deck (Fig. 21e and f). Compared with the foldable flap wingsail, the foldable traditional wingsail is lighter in weight and has a more reliable structure; however, the foldable traditional wingsail has a large area outside of the hull after being folded, and the friction resistance is larger when navigating underwater.

Seagull autonomous sailboat, developed by the Shenyang Institute of Automation, is also equipped with a foldable wingsail. From Fig. 22, it can be observed that the top half of the sail is rigid. For the bottom half of the sail, the leading edge is rigid, but the trailing edge is flexible. The top half of the sail can rotate on its axis.

Other autonomous sailboats with foldable wingsails include the Bluebottle and Petrel-dual (see Table 9). Unlike the Triton, Bluebottle and Petrel-dual use foldable arc-shaped wingsails. The advantage of this design is that the sailboat does not need to be slotted in the deck to accommodate toppled wingsails.

5. Multi-sail layout and new types of sails

5.1. Multi-sail layout of autonomous sailboat

The sails introduced in section 4 are all single sail. Autonomous

sailboat can use two sails replace single sail. The advantages of the method are as follows.

- 1) As shown in Fig. 23, using two smaller sails instead of one large sail can lower the CE of the sail. This adjustment reduces the heeling moment and allows for a lighter and shallower keel.
- 2) The double-sail configuration can serve as a substitute for the rudder if it becomes damaged, enhancing the boat's survivability in harsh sea conditions.

As shown in Table 10, the ARRTOO autonomous sailboat is equipped with flexible double sails, while the A-Tirma G2, ARC, Force 12 Xplorer, and SJTU-Seastel autonomous sailboats are equipped with rigid double wingsails.

Unlike a single wingsail, interference occurs between two sails, with the aft sail affected by the wake from the fore sail. To minimize this interference, it is essential to research the aerodynamic performance of double sails and the factors influencing it. Key factors affecting the overall aerodynamic performance of double sails include the distance between the sails, the chord length ratio, the selection of airfoils for the fore and aft sails, and the angle of attack, as shown in Fig. 24.

Sun et al. [117] used the vortex panel method to analyze the lift coefficient of rigid double wingsails. Their results indicated that, within an AOA range of 1° – 11° , increasing the distance between the aft sail and the fore sail reduces the influence of the fore sail on the aerodynamic performance of the aft sail. Additionally, a thinner fore sail results in less interference with the aft sail's aerodynamic performance. Sun et al. [118] also found that the total lift coefficient increases with the chord length of the fore sail.

However, references [117,118] have limitations as they were based on 2D analysis and did not account for viscous boundary layer corrections or 3D effects. Future studies should incorporate boundary layer theory to explore the aerodynamic characteristics of 3D wingsails in greater detail.

5.2. New sails of autonomous sailboats

The skysail, also known as a kite sail, is illustrated in Fig. 25. The skysail system used by large sail-assisted ships comprises a skysail, control rope, control pod, towing rope, and other equipment connected to the ship through a force transmission point [119]. The world's first skysail-driven cargo ship was launched in 2007 and was equipped with a 160 m^2 Beluga skysail. The sail area can be expanded to 320 m^2 to achieve greater traction [120].

Similarly, HIEJIMA proposed a scheme to drive an autonomous sailboat using a skysail [122], as shown in Fig. 26. In this scheme, the sailboat is towed by the wind and powered by underwater turbines.

The skysail operates at high altitudes of 100–300 m, where it can harness higher-speed winds. With the same sail area, the driving force generated by a skysail is six times greater than that of a traditional sail. Additionally, the heeling moment exerted by the skysail on the ship is less than that of a traditional sail. However, skysails have some drawbacks that make them less suitable for deployment on autonomous sailboats.

- 1) Skysail is composed of flexible structures. The sudden change in wind direction causes the skysail to deform and lose its power.
- 2) When boat sailing against the wind or reaches a certain speed, the thrust of the sail is reduced [123].
- 3) It is difficult to control a skysail system automatically.
- 4) The skysail system is large. Compared with large ships, autonomous sailboats are smaller and have difficulty carrying skysail systems.

6. Discussion

Compared with sailing yachts, autonomous sailboats face several

Table 10
Autonomous sailboats with double wingsails.

Platform name	Research institution	Reference	Parameter(s)	Image
ARRTOO	Aberystwyth University	[113]	Length: 2 m, Displacement: 29.5 kg	
A-Tirma G2	University of Las Palmas De Gran Canaria	[112]	Length: 2 m, Wingsail airfoil: NACA0009, Wingsail height: 1.05 m, Wingsail average chord length: 0.225 m, Wingsail material: carbon fiber	
ARC	Aberystwyth University	[114]	Wingsail height: 1.07 m, Wingsail width: 0.2 m	
Force 12 xplorer	Open Ocean Robotics	[115]	Length: 2.5 m, Displacement: 50 kg	
SJTU-Seastel	Shanghai Jiao Tong University	[116]	Length: 1.5 m, Wingsail airfoil: NACA 0012	

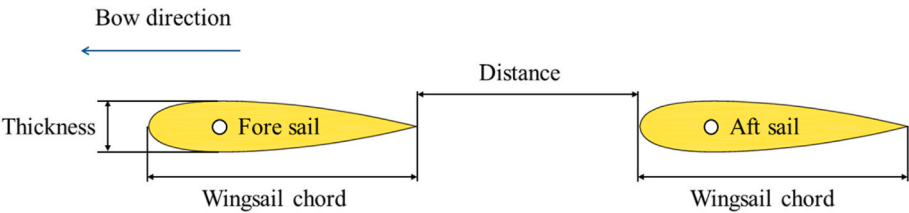


Fig. 24. Rigid double wingsail schematic diagram.

limitations in sail usage, such as the following.

- 1) Unlike sailing yachts, autonomous sailboats cannot easily switch between different sails based on working conditions, and the sail area of rigid wingsails is generally difficult to adjust.
- 2) In sailing yachts, the righting moment can be adjusted by changing the crew's position as ballast on the port or starboard side. However, in autonomous sailboats, the maximum righting moment is fixed, which impacts the determination of the maximum sail area.
- 3) The sails of autonomous sailboats are controlled automatically; therefore, the complex control methods used in sailing yachts are no longer applicable.
- 4) Given the limitations of electric energy supply, it is essential to study sail adjustment methods that minimize power consumption and facilitate automatic control. In general, the development of

autonomous sailboat sails has followed the process depicted in Fig. 27. Building on the traditional wingsail, the flap wingsail and self-trimming wingsail were developed to enhance aerodynamic efficiency and reduce power consumption. Rigid wingsails have become popular in autonomous sailboats. The basic configuration of a rigid wingsail can be further optimized by adapting it to various working conditions to improve its performance.

According to the current technical development and future potential applications of autonomous sailboats, the following aspects may be key topics for future research.

High strength and light weight: Sails are the core power sources for autonomous sailboats, and damage to the sails can result in a loss of driving force. Sails near the sea surface are vulnerable to waves, and the wingsail structure is subjected to alternating loads from wind and

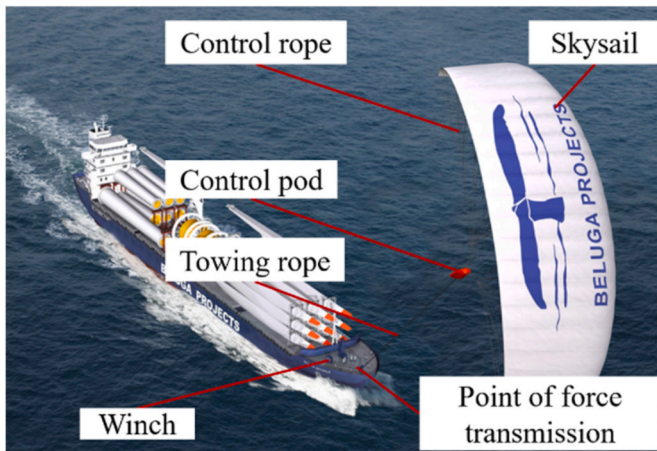


Fig. 25. Skysail system [121].

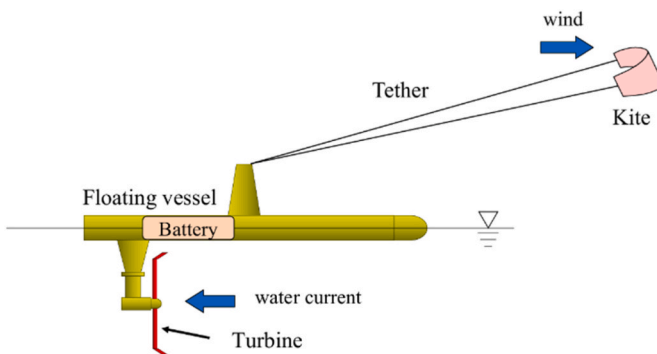


Fig. 26. Schematic diagram of an autonomous sailboat with skysail installed [122].

sailboat rolling during long-term navigation, making it prone to fatigue damage. Fig. 28 illustrates the wingsail structures of SubSeaSail [124], WASP [125], Petrel-Dual [97] and Seagull, along with the types of loads and corresponding failure types.

Improving the overall strength and reliability of wingsails often results in increased weight, which raises the center of gravity of the autonomous sailboat and reduces lateral stability. Therefore, achieving a lightweight design for high-strength wingsails is crucial. Research could focus on optimizing the carbon fiber ply of the wingsail skin, the material of the wingsail skin, the number of ribs, and the cross-sectional shape of the mast.

Multifunctional Integration: Wingsails primarily provide driving force for autonomous sailboats. When sailing upwind, the wind approaches from the bow, and any antennas or components installed in this area can affect the airflow to the wingsail. For instance, antennas, as

shown in Fig. 29, can reduce the driving force generated by the wingsail. Installing antennas at higher locations can improve communication, so finding an optimal balance between wingsails and antennas is important. So, there is a good combination point between wingsails and antennas.

With the growing use of autonomous sailboats for scientific expeditions and the need to install scientific payloads at specific heights, more equipment such as weather stations and cameras will be mounted on wingsails. The large area of the wingsail also makes it suitable for installing solar panels, which can effectively complement the electric power supply.

In the future, wingsails are expected to evolve from simple components into systems that integrate energy, communication, and sensing capabilities, as shown in Fig. 30. However, this integration presents three critical problems that need to be addressed.

- 1) The installation of scientific payloads on the wingsail can impact aerodynamic performance. To mitigate this, sensors and antennas can be concealed within the wingsail, with only the probes exposed. Antennas can be housed in shells made of wave-transparent materials to minimize wind resistance. Therefore, it is important to study the conformal design of scientific payloads and the wingsail, and to optimize the internal space of the wingsail.
- 2) When different antennas are installed on a wingsail, it is crucial to filter out interference between adjacent frequency bands and ensure the effective coexistence of the antennas.
- 3) Since the wingsail is rotationally connected to the hull, it is important to ensure the stability of both electrical and signal connections. An electrical slip ring can be used to maintain reliable connections for power and signals between the wingsail and hull.

Deformable and adaptive: Wingsails are situated in the lower boundary layer of the atmosphere, where wind speed changes rapidly. Wind shear refers to changes in true wind direction with altitude, while wind gradient denotes changes in true wind speed with altitude. Together, these factors influence the wind acting on the sails and can significantly impact sail trimming. Fig. 31 illustrates the flow with speed gradient and twist affecting the wingsail. A wingsail equipped with a flap can adjust aerodynamic parameters by deflecting the flap. However, mainsail and flap are typically constructed from composite materials, such as carbon fiber or glass fiber, resulting in a rigid sail surface. Since the wingsail is a non-twisting rigid body, variations in wind velocity gradient and wind shear create different apparent wind angles at various heights. Consequently, while the bottom of the wingsail may be at an optimal AOA, the top may deviate from this optimal angle and enter a stall state, thereby affecting overall aerodynamic efficiency.

With advancements in morphing wings and miniature pressure sensing technologies, morphing wing technology can be integrated into the wingsails of autonomous sailboats. As shown in Fig. 32, an autonomous sailboat can determine the appropriate airfoil based on wind changes, and then control the driving mechanism within the wingsail.

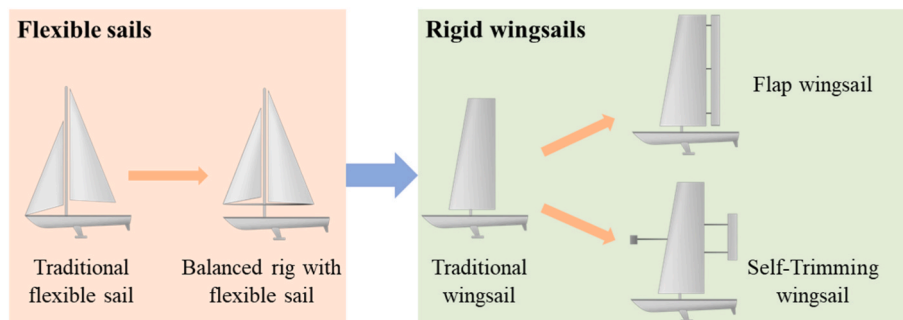
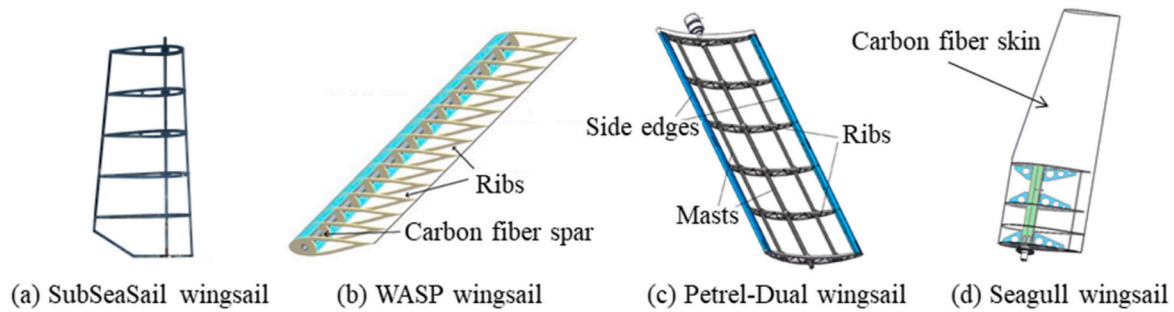


Fig. 27. Development roadmap of sails of autonomous sailboats.



Loads type	■ Aerodynamic loads	■ Structural loads	■ Dynamic loads
Force source	Aerodynamic force	Mass force	Inertial force
Load position	Wingsail skin ↓ wingsail rib	Mast	Mast
Failure types	Structure instability		Fatigue failure

Fig. 28. The loads type, force source, loads position, and failure types of wingsails.



Fig. 29. Antennas and other components installed on the Seagull autonomous sailboat.

This allows for precise and controllable deformation of the windward side of the wingsail skin, thus modifying the wingsail's airfoil to achieve optimal driving force. For further research into morphing wingsails, considerations should include the materials used for the envelope, actuators, and both internal and external structural design of the wingsail.

Interchangeable and modular: Autonomous sailboats must operate under various conditions. Given the cost and limited deck area of scientific research ships, a good solution is to match a hull with different sails and to combine them based on specific sailing requirements. Therefore, the electronic and physical interfaces between the wingsails and the hull should be studied with respect to interchangeability to enable quick changes of the wingsails. The scheme of one hull plus n wingsails is as shown in Fig. 33. Critical aspects to address include analyzing how external environmental disturbances affect the safety and operational state of the wingsail structure under different conditions, and designing a wingsail configuration that is adaptable to various working conditions.

Self-diagnosis and intelligence: When an autonomous sailboat sailing at sea surface for extended periods, the sails endure harsh

conditions and are prone to damage. Damage to the integrity of the wingsail directly impacts its aerodynamic characteristics. The types of wingsail failures are listed in Table 11.

Autonomous sailboats should incorporate self-diagnosis technology to monitor sail integrity. Strain gauges, optical fiber sensors, and other sensors within the sail can detect damage and provide real-time diagnostics of the sail's condition. By using features observed after a failure as prior knowledge, the control system can determine whether the sailboat can continue its mission. An intelligent sail control algorithm should automatically adjust relevant parameters following damage to maintain optimal energy consumption and control accuracy.

High performance bionics: Bionics involves drawing inspiration from natural biological systems to drive design innovations. Problems faced by sailboats can be addressed by mimicking animal structures. For instance, the internal cavity of the wingsail can replicate the honeycomb structure found in bones to enhance overall strength while reducing weight. Folding wingsails can be designed to retract similarly to fins. Additionally, the wingsail's cross-section can be modeled after the wings of gliding birds (see Fig. 34), such as albatrosses, to use wind energy more efficiently and generate greater lift.

7. Conclusion

As a new type of mobile observation platform, autonomous sailboats are crucial for acquiring long-term, large-scale air-sea interface field observation data. The performance of the sails, which are the core devices for converting wind energy into driving force, significantly impacts the sailing performance and safety of these sailboats.

To date, autonomous sailboat sails have evolved through three stages: traditional flexible sails, balanced rig with flexible sails, and rigid wingsails. Advances in wingsail technology have led to the development of various configurations, including wingsails with flaps or tails. Rigid wingsails represent the future direction for autonomous sailboat sails. However, further research is needed to enhance their aerodynamic efficiency, structural safety, and reliability. Key areas of focus include high strength, lightweight construction, multifunctional integration, interchangeable modularization, self-diagnostic intelligence, and deformable adaptability. Improving the navigation performance of autonomous sailboats and enabling year-round operation will depend on these advancements.

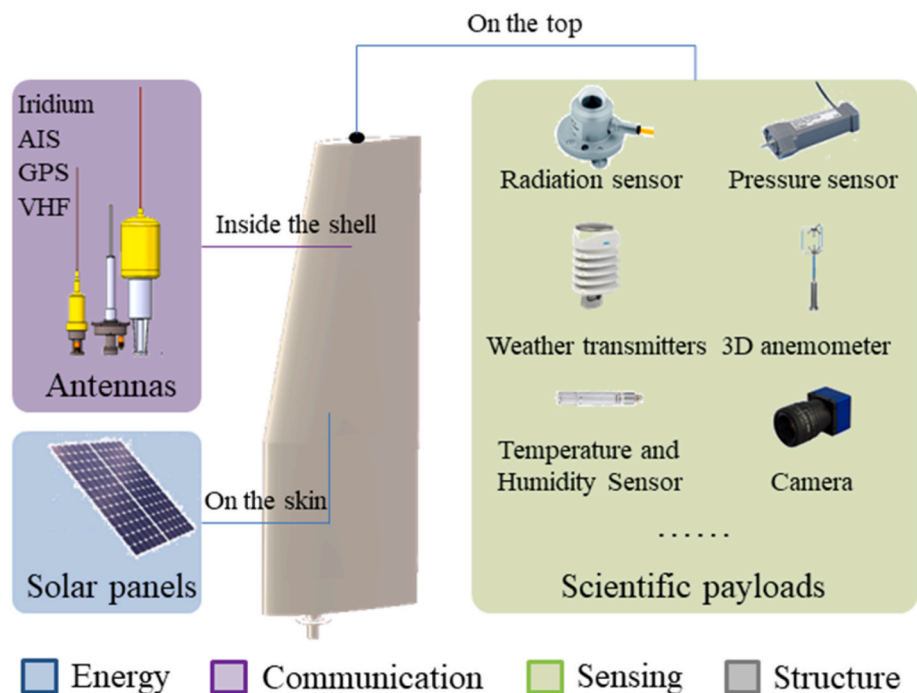


Fig. 30. Wingsail with multifunctional integration.

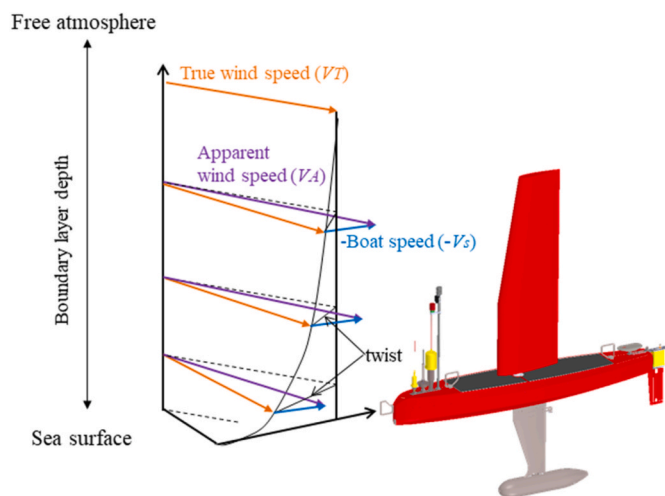


Fig. 31. Flow with speed gradient and twist onto wingsail.

CFD and wind tunnel testing are the primary methods used to study wingsail aerodynamic performance. With the advancement of artificial intelligence, technologies such as machine learning could be integrated to further enhance the study of wingsail aerodynamics.

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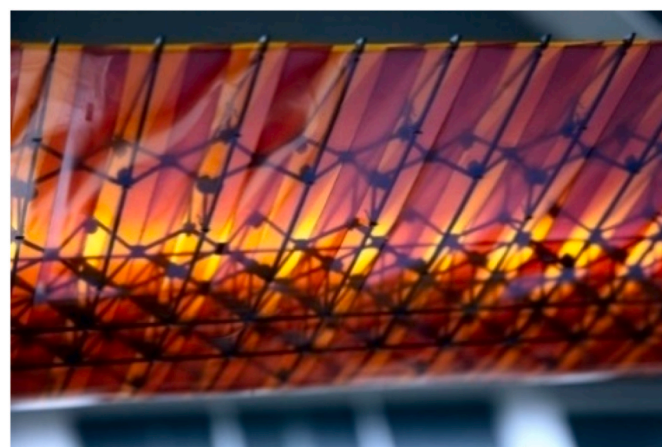


Fig. 32. Morphing wing [126].

CRediT authorship contribution statement

Zhaoyang Sun: Conceptualization, Writing – original draft, Funding acquisition, Investigation, Visualization. **Ao Feng:** Writing – original draft, Investigation. **Jiancheng Yu:** Supervision, Project administration. **Wentao Zhao:** Writing – review & editing. **Yan Huang:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.



Fig. 33. Modular wingsail.

Table 11
Autonomous sailboats wingsails failure types.

Failure modes	Wingsail skin depression	Wingsail skin damage	Wingsail skin fracture	Restricted mast rotation angle	Mast cannot rotate	Mast broke, wingsail lost
Diagrams						
Causes	Seawater immersion, Wind load	Wave impact load	Wave impact load	Transmission system failure	Transmission system failure	Mast fatigue fracture
Effects	Driving force decreased (Low), inconsistent sailing capabilities on port and starboard	Driving force decreased (Mid), inconsistent sailing capabilities on port and starboard	Driving force decreased (High), inconsistent sailing capabilities on port and starboard	Limited sailing ability	Sailing under specific condition	Driving force lost
Corresponding Strategy	Online identification and adjustment of the optimal wingsail AOA control model	Online identification and adjustment of the optimal wingsail AOA control model	Online identification and adjustment of the optimal wingsail AOA control model	Adjust the path planning strategy, Waiting for rescue	Waiting for rescue	Waiting for rescue

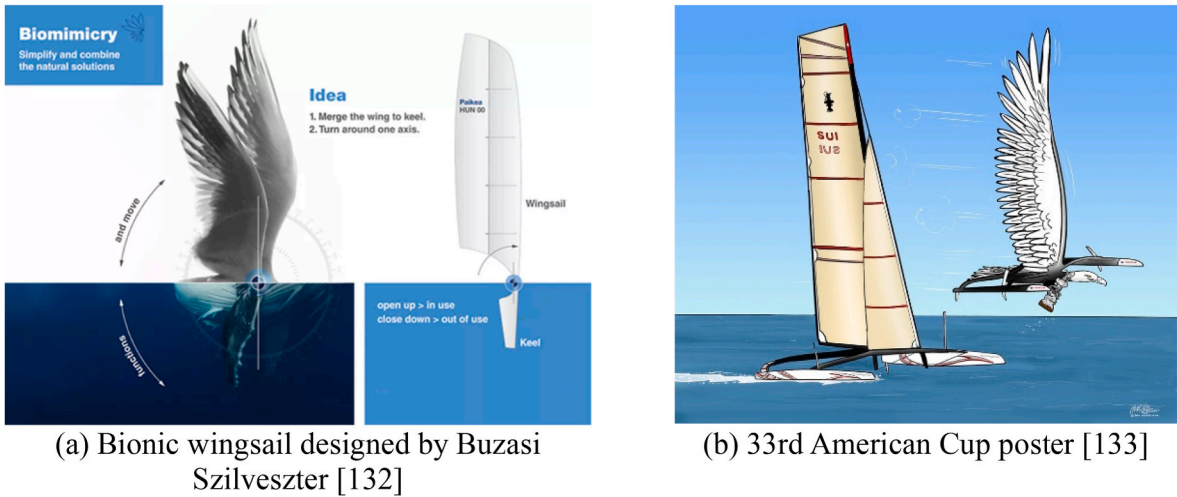


Fig. 34. Bionic wing schematic.

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